A major roadblock is the difficulty of getting direct experimental data. Gravity is a very weak force; the electric repulsion between two electrons is about 10^{43} times stronger than their gravitational attraction. In any realistic experiment involving just a few particles, where we might expect quantum effects to be visible, the force of gravity is utterly negligible compared to other influences. We can imagine building a particle accelerator powerful enough to smash particles together at the Planck energy, where quantum gravity should become important. Unfortunately, if we simply scale up the technology in current machines, the resulting accelerator would have to be light-years in diameter. It's not a feasible construction project at this time.

There are also technical problems with the theory itself, in addition to the conceptual ones just mentioned. General relativity is a classical field theory. The field involved is called the *metric*. (The symbol $g_{\mu\nu}$ in the middle of Einstein's equation represents the metric, and the other quantities depend on it.) The word "metric" ultimately derives from the Greek *metron*, "something used to measure," and that's exactly what the metric field allows us to do. Given a path through space-time, the metric tells us the distance along that path. The metric essentially updates Pythagoras's theorem, which works in flat Euclidean geometry but has to be generalized when spacetime is curved. Knowing the length of every curve suffices to fix the geometry of spacetime at every point.

Spacetime has a metric even in special relativity, or for that matter in Newtonian physics. But that metric is rigid, unchanging, and flat—the curvature of spacetime is zero at every point. The big insight of general relativity was to make the metric field into something that is dynamical and affected by matter and energy. We can attempt to quantize that field just as we would any other. Small ripples in the quantized gravitational field look like particles called *gravitons*, just like ripples in the electromagnetic field look like photons. Nobody has ever detected a graviton, and it's possible that nobody ever will, since the gravitational force is so incredibly weak. But if we accept the basic principles of general relativity and quantum mechanics, the existence of gravitons is inevitable.

We can then ask what happens when gravitons scatter off each other or off other particles. Sadly, what we find is that the theory predicts nonsense, if it predicts anything at all. An infinite number of input parameters are needed to calculate any particular quantity of interest, so the theory has no predictive power. We can restrict our attention to an "effective" field theory